

Direction of Arrival Estimation In Wireless Communication Based on Deep Neural Networks

Fatemeh Lotfi

University of Colorado, Colorado Springs
Colorado Springs, CO 80918
flotfi@uccs.edu

Abstract

Emerging applications require seamless network coverage. One of the promising technologies that provide seamless coverage is super directive transmission (contrary to broadcasting communication). In this regard, knowing the Direction of Arrival (DoA) is one of the vital prerequisites. In this paper, we develop a Deep Neural Network (DNN) that learns to estimate the DoA of the received signals. We attempt to do this for various fields of view and several distinct DoAs, both in one and two signals spaces. Furthermore, the results show the effect of angular range and number of antenna elements on the DoA estimation accuracy. In the two signals spaces, we also compare our DNN results for different Signal-to-Noise Ratios (SNRs) with one classical method including, Multiple Signal Identification (MUSIC) and Cramer-Rao Lower Bound (CRLB). The simulation results indicate that the proposed DNN approach predict DoA with up to 60% greater precision than the MUSIC algorithm and can attain the CRLB.

Introduction

Researchers have considered Direction of Arrival (DoA) estimation for decades because of its various usage including, tracking, localization, and gesture recognition. One requirement of DoA estimation is antenna array, which consists of separate components of the antennas arranged in a particular mapping. Some applications, such as vehicles, ships, aircraft, and base stations, use these kinds of antennas. They play a crucial role in increasing the capability of channels. Due to their ability to redirect nulls and beams to minimize interference and maximize gain in particular directions, as well as their ability to predict the angle of arrival of the received signal, they are attractive for mobile wireless communications. DoA estimation has a long history and, there are a lot of techniques and algorithms. Improving accuracy and super-resolution, improving adaptation to challenging scenarios with minimal snapshots, low signal-to-noise ratio (SNR) are hot topics in DoA estimation (Guo et al. 2020; Mecklenbräuker et al. 2020; Zhou et al. 2020; Papageorgiou, Sellathurai, and Eldar 2020; Zhu et al. 2020; Chen, Gong, and Mao 2020). Parametric

classical methods fall into categories, subspace-based approaches (Gupta, Aditya, and Datta 2016), Multiple Signal Classification (MUSIC), and Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT), beamforming techniques (Krishnaveni, Kesavamurthy, and Aparna 2013) and, maximum likelihood methods (Mecklenbräuker and Gerstoft 2019). These techniques are based on mappings into array outputs from signal directions and vice versa. The accuracy of these methods depends on the mapping accuracy. However, since most system defections may be due to non-ideal sensor design and development, array assembly, interference, and background radiation are non-ideal.

Recently, researchers have started using frameworks equipped with machine learning techniques for estimating the signal's DoA. These methods are data-driven and do not depend on ideal array assumptions. These kinds of approaches create training data sets with DoA labels and then, using machine learning techniques, extract a mapping between output and signal direction. In our work, leveraging deep neural network techniques, we propose and implement a method for DoA estimation. Our objective is to increase the predictive accuracy of DoA compared to other powerful classical and theoretical methods like MUSIC and Cramer-Rao Lower Bound (CRLB). Moreover, we compare the performance of the proposed method in both noisy and noiseless environments.

Related Work

There are numerous works in the literature using machine learning algorithms for DoA estimation. In this paper, we investigate the literature based on deep neural networks for DoA estimation.

One of the most recent promising approaches for DoA prediction is the neural networks and deep learning algorithms, which recently attracted a lot of attention (Takeda and Komatani 2016; Adavanne, Politis, and Virtanen 2018). These papers develop a deep learning method for DoA estimation in microphone array and use time-frequency images of signals as input. In both of these papers, source's angle resolution is rough, 5° (Takeda and Komatani 2016) and 10° (Adavanne, Politis, and Virtanen 2018). In (Chen, Gong, and Mao 2020), the authors describe a method that integrates detection with Deep Neural Network (DNN) based DoA estima-

tion. They split the DoA estimation network into multiple DNNs to minimize the necessary training data. Finally, they run an offline training phase that decreases the computation time for signal direction estimation. Also, in (Baig and Malik 2013) the authors provide comprehensive research on array processing and describe various kinds of beamforming methods. They did it along with their advantage in communication systems. Moreover, they illustrate different DoA estimation methods including, spectral estimation, Minimum Variance Distortion less Response (MVDR) estimator, linear prediction, maximum entropy, Maximum Likelihood (ML), MUSIC, ESPRIT, and compare their performance.

In (Papageorgiou, Sellathurai, and Eldar 2020), the authors implement a deep Convolutional Neural Network (CNN) for DoA estimation in a low SNR regime. They consider DoA estimation as a multi-label classification problem and, convolutional layers extract the features from multi-channel input data. Subsequently, their proposed method can infer the number of sources. Then the fully connected layers conclude the DoA by using the convolutional layers' output. Based on Root Mean Square Error (RMSE), the performance of their method is well for the lower SNR regime. Also, their angle estimation is sufficiently good within the 1° resolution.

In (Zhu et al. 2020), the authors propose ensemble learning to reduce the processing time. They use five different CNNs independently to determine azimuth and elevation angles. To perform a 2D DoA estimation, they consider a Uniform Circle Array (UCA) in their model. Furthermore, in each CNN, CNN's input is the spatial covariance matrix of the received signals, and their output results in the final prediction for azimuth and elevation angles. Also, their method can immediately estimate DoA that is suitable for real-time applications. In (Liu, Zhang, and Philip 2018), the authors use the DNN method to increase the accuracy of DoA estimation in practical sensor arrays. Furthermore, they consider the array imperfection effect in their model. Then use preprocessing methods like spatial filters to reduce the distribution divergence of input data. To this end, their DNN framework consists of a multi-task autoencoder as preprocess filter to denoise the DNN input and some parallel multi-layer classifier. Finally, they compare the proposed model with SVR-based and MUSIC methods to show their performance is comparable to other methods, like MUSIC.

Also, authors in (Kase et al. 2018) propose a DNN with four layers fully connected (FC) to estimated the DoA of two targets using the signal covariance matrix. In (Wu, Liu, and Huang 2019), authors introduce a deep convolution network framework with four convolutional layers that could efficiently obtain DOA estimates in near real-time.

Data

To feed the model with data, we require a large amount of labeled data to train the DNN. We inputted our model with the dataset produced through simulation in a MATLAB environment. It is the covariance matrix of received signals. To do this, we randomly generate a large amount of data with the simulation of uniform linear array antenna and different target numbers, target SNR, and target directions in noisy conditions. Then we calculate the received signal covari-

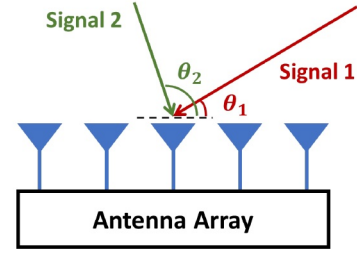


Figure 1: Uniform Linear Array (ULA)

ance matrix as the feature matrix. The Algorithm 1 shows the dataset generation's steps. Indeed, Table 1 depicts the simulation parameters. We consider 7×10^4 independent samples of one signal with different random directions in $\{-\pi/2 : \pi/2\}$ and noisy data with various SNRs ranging from 0 dB to 10 dB with 2 dB increments as dataset I. Finally, we generate 7×10^4 different received signal covariance matrix for each SNR.

Algorithm 1: Dataset generation steps

Result: $D = \{R_k, \theta_k\}, k = 1, \dots, 70K$
for Different values of SNR **do**
 for Number of targets in environments **do**
 Step 1: Generate a set of random DoAs as the target k-th's directions.
 Step 2: Calculate the received signal with SNR i-th in linear array elements with different phase.
 Step 3: Calculate the covariance matrix of received signal ($[R] : N \times N$)
 end
 Step 4: Generate the dataset by real, imaginary and angle values of complex elements of covariance matrix. ($[D] : N \times N \times 3$)
end

The proposed DNN's input data is a real-valued $N \times N \times 3$ matrix, with the third dimension representing various channels. The first and second channels correspond to the real and imaginary portions of R_y , respectively, while the third channel corresponds to phase entries, $\angle\{R_y\}$.

Also, we generate another dataset for two different signals with random directions and different SNR in a single communication space as dataset II.

Problem formulation

In this work, we consider a Uniform Linear Array (ULA) antenna with N elements, Fig. 1, and assume the signal comes from far-field¹. By considering d as the distance between the elements, we can define the received signal vector at the

¹With the far-field assumption, the wavelength is much less than the distance from the source to the antenna.

Table 1: Simulation Parameters

Notation	Parameter	Value
N	Number of array elements	10
d	Distance between elements	$\frac{\lambda}{2}$
M	Number of snapshots	7×10^4
K	Number of targets	$\{1, 2\}$
f_s	Target signal carrier frequency	$500 MHz$
γ	Signal-to-Noise Ratio (SNR)	$\{0, 2, 4, 6, 8, 10\} dB$
θ	Target direction range	$\{-\frac{\pi}{2} : \frac{\pi}{2}\}$

output of the N -element antenna with K far-field source as:

$$\begin{aligned}
 \mathbf{y}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{n}(t), \\
 \mathbf{y}(t) &= [y_1(t), y_2(t), \dots, y_N(t)]^T, \\
 \mathbf{x}(t) &= [x_1(t), x_2(t), \dots, x_K(t)]^T, \\
 \mathbf{n}(t) &= [n_1(t), n_2(t), \dots, n_N(t)]^T, \\
 \mathbf{A} &= [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_K)],
 \end{aligned} \tag{1}$$

where, $\mathbf{y}(t)$ is the received signal vector, $\mathbf{x}(t)$ is the source signal vector from far-field, and $\mathbf{n}(t)$ is the white Gaussian noise vector. $\mathbf{A}$ is the steering vector towards the direction of arrival of K signal, and the column of $\mathbf{A}$ written as:

$$\mathbf{a}(\theta_k) = [1, e^{\frac{2\pi d}{\lambda} \sin(\theta_k)}, \dots, e^{\frac{2\pi d}{\lambda} \sin(\theta_k)(N-1)}]^T, \tag{2}$$

where $\lambda = \frac{c}{f}$, f , c , and T is the wavelength of the received signal, carrier frequency, the speed of light, and transpose operator, respectively. From (1), the covariance matrix $\mathbf{R}$ of the received signal shown as follows:

$$\begin{aligned}
 \mathbf{R}_y &= E[\mathbf{y}(t)\mathbf{y}^H(t)] \\
 &= \mathbf{A}\mathbf{R}_s\mathbf{A}^H + \sigma^2\mathbf{I},
 \end{aligned} \tag{3}$$

where E is the statistical expectation operator, H is the conjugate transpose operator, and the covariance of the signal is known as the signal's power. Since $\mathbf{R}_y$ is a complex-valued matrix, we consider real, imaginary, and phase values of the covariance matrix as an input value. Thus the number of inputs in DNN would be $N \times N \times 3$, and they are the real values.

Model approach

One of the powerful methods for DoA estimation is neural networks. A neural network consists of an input layer, several hidden layers, and an output layer. In each layer neurons with a weighted summation (w_i) and a nonlinear activation function (f), transfer their output (y_l) to the next layer according to following equation:

$$y_l = f\left(\sum_{i=1}^N x_i w_i\right). \tag{4}$$

The activation function f is heuristic and will be selected based on the problem. During the training epochs, using

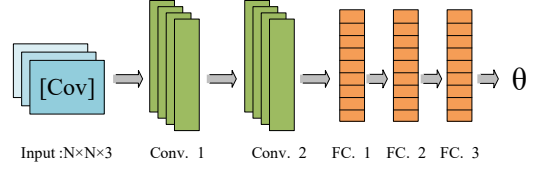


Figure 2: Architecture of a multiple-layered deep neural network.

the back-propagation technique, the network updates its weights. Depending on the number of hidden layers in the network, the network falls into shallow or deep categories. We can elaborate our approach as follows:

$$\hat{\theta} = f_{NN}(\mathbf{R}_{xx}), \tag{5}$$

where, $\mathbf{R}_{xx}$ is the covariance matrix of received signals, and f_{NN} is an approximation function created with neural networks similar to Fig. 2.

Model implementation and Results

We design and implement a multiple-layered deep neural network using Tensorflow and Keras libraries. The codes are running on the Google Colab platform. To consider the best parameters for the model, we test different architectures with different depths to find a model with sufficient predictability. To do this, we consider a neural network with five different hidden layers consist of two convolutional layers with $\{128, 64\}$ channels and the kernel size $\{3, 2\}$. The activation function at the output of each hidden layer is the rectified linear unit (ReLU) function, $ReLU(x) = \max(0, x)$. Also, because of the type of input data, it is better not to use pooling layers. Empirical results show it would result in loss of information and poor performance. After the Convolutional layers, a flatten layer transfers the learning process to fully connected (FC) layers. Then, there are three FC layers with $\{128, 64, 32\}$ neurons. From the DSP (Digital Signal Processing) point of view, each neuron resembles matched filter functionality. It means, in the first layer of fully connected, we leverage 64 different matched filters to distinguish signal features extracted by CNN. Then, we load in results of the first layer matched filter to the second 64, and then the final 32 consecutive matched filters and extract more profound features. The features enable us to estimate the target DOA. In the process, the epoch and batch size is 100 and 64, respectively. We consider the $ReLU$ function as a nonlinear activation function and the $Adam$ optimizer as an exploration mechanism of training data. Fig 2 depicts proposed architecture. We have considered this architecture inspired by (Papageorgiou, Sellathurai, and Eldar 2020). Table 2 shows some other implementation settings.

In the rest of the paper, we consider two scenarios, with one and two targets in space. Then we estimate the target's DoA for each scheme.

Prediction of DoA for one signal space

In this section, we consider one target scenario and predict the target DoA with the proposed network. Fig. 3 shows the

Table 2: Implemented Neural network setting

Setting	Value
Number of testing samples	10^4
Number of training samples	6×10^4
Batch size	64
Number of epochs	100
Hidden layers (two Conv. layers)	[128, 64]
Kernel size (two Conv. layers)	[3, 2]
Activation function	<i>ReLU</i>
Hidden layers (three FC. layers)	[128, 64, 32]
Optimizer	<i>Adam</i>
Loss function	<i>mse</i>

training model loss for object with SNR=10 dB in different epochs. As we see, the implemented NN properly converged.

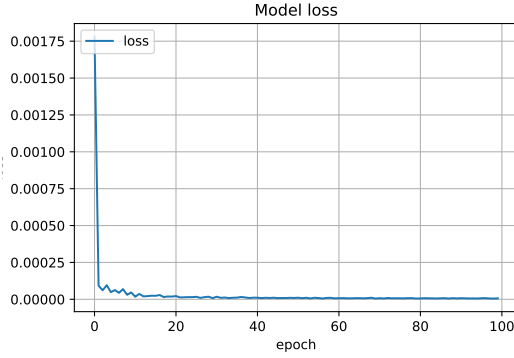


Figure 3: Training loss with respect to number of epoch for one target in noisy environment with SNR=10 dB.

Also, Fig. 4 shows the predicted DoA versus the actual DoA for target SNR=10 dB. As we know, predicted DoA at the boundaries of the full-angular range, $[-\pi/2 : \pi/2]$, has a poor performance. It happens because the received signals from targets in the boundary angles would not receive properly. That is why most of the literature did not consider the full range angles for DoA prediction. In this part, we would consider 80% of the full-angular range for target DoA. Also, in the following, we discuss the effect of this parameter on the DoA estimation accuracy.

Prediction of DoA for two signals space

In this section, we consider two targets in space and predict the DoA.

Fig. 5 shows the training model loss for the object with SNR=10 dB in different epochs. Also, Fig. 6 shows the predicted DoA versus the actual DoA for target SNR=10 dB in 80% of full angular range for target DoA.

Performance Evaluation

To test the performance of our proposed DNN model, we consider some popular methods to compare their performance with our DNN method. Furthermore, we leverage two

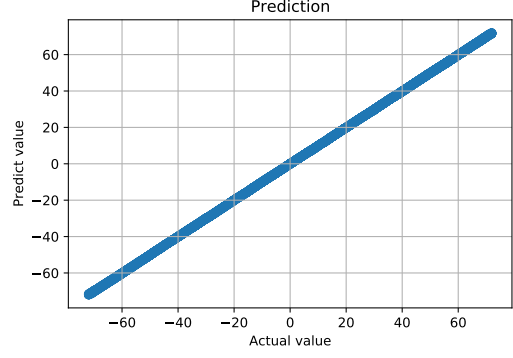


Figure 4: Predicted DoA vs. actual DoA for one target in noisy environment with SNR=10 dB.

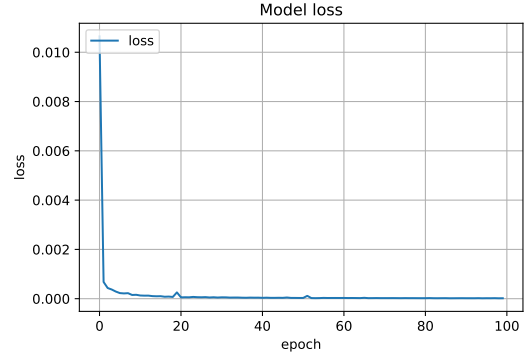


Figure 5: Training loss with respect to number of epoch for two targets in noisy environment with SNR=10 dB.

evaluation metrics RMSE and R-squared show the performance of our model. Table 3 and Table 4 show the evaluation metrics for different SNR in one and two signals space, respectively.

Table 3: Evaluation metrics for one signal space

SNR	RMSE	R^2
0	0.4	0.92
2	0.37	0.92
4	0.35	0.94
6	0.32	0.97
8	0.28	1
10	0.25	1

Table 4: Evaluation metrics for two signals space

SNR	RMSE	R^2
0	0.47	0.9
2	0.4	0.92
4	0.35	0.94
6	0.32	0.96
8	0.29	1
10	0.26	1

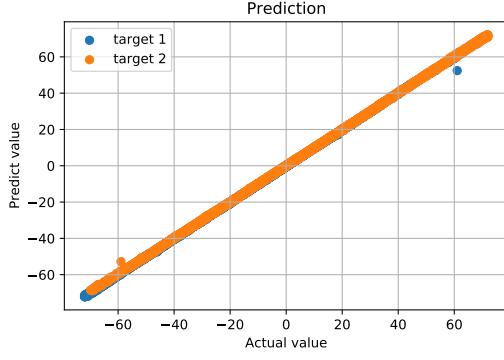


Figure 6: Predicted DoA vs. actual DoA for two targets in noisy environment with SNR=10 dB.

The effect of different angular ranges on DoA estimation

As mentioned earlier, the target angular range would impact estimating accuracy. In this part, we want to consider its effect. Table 5 shows the RMSE for different target angular range 100%, 90% and 80% of range $[-\pi/2, \pi/2]$ for target direction. According to the results in Table 5, by choosing a smaller angular range, we will achieve higher angular accuracy. Indeed, achieving higher accuracy also has a cost. We have compromised received signals with angles out of angular range with more accuracy. However, it is a trade-off and depends on the application.

Table 5: RMSE of different angular range

SNR	Range 100%	Range 90%	Range 80%
0	20.11	0.85	0.47
2	19.42	0.8	0.4
4	18.74	0.68	0.35
6	17.4	0.62	0.32
8	16.29	0.53	0.29
10	14.42	0.4	0.26

The effect of the number of antenna elements on DoA estimation

The number of antenna elements in the array would impact the received signal quality from the target. Hence, this is a parameter that would influence the angular prediction accuracy. In Table 6, we show the impact of the number of antenna elements on DoA estimation accuracy in the full-angular range. As we can see in Table 6, an antenna array with more elements has less RMSE. The choice of the number of antenna elements depends on the application and location of the antenna.

Performance comparison with classical methods MUSIC and CRLB

In this section, we would compare the proposed DNN approach to the approved method, MUSIC. Then we investi-

Table 6: RMSE of different number of antenna elements

SNR	Num. of elements = 10	Num. of elements = 4
0	20.11	27.6
2	19.42	25.7
4	18.74	24.8
6	17.4	23.26
8	16.29	22.6
10	14.42	20.93

gate how close their results would be to Cramér-Rao lower bound (CRLB). In the following, we briefly explain these methods.

• Cramer-Rao Lower Bound (CRLB)

As we know, the DoA estimators want to find the direction of signal from the received noisy signal. The Cramer-Rao Lower Bound (CRLB) is a theoretical method that gives a lower bound for the variance of θ as follows:

$$\text{Var}[\hat{\theta}] \geq -E\left[\frac{\partial^2}{\partial \theta^2} \ln f_y(\theta)\right]^{-1}, \quad (6)$$

where, $f_y(\theta)$ is pdf of the received signal. Assuming the Gaussian noise in (1), $f_y(\theta)$ is recognizable and this approach is achievable. The CRLB method states that the estimators which achieve this lower bound would be efficient and reaches the lowest mean square error. This approach has better performance if we have an antenna array with many elements.

When there are multiple signals in the environment, there is a lower bound for the estimated DoA's variance by the CRLB as follows:

$$\text{Var}[\hat{\theta}] \geq \frac{\sigma_n^2}{2n} \left(\text{Re} \left[\mathbf{H} \mathbf{O} (\mathbf{A}^H \mathbf{R}^{-1} \mathbf{A})^T \right] \right)^{-1}, \quad (7)$$

where,

$$\mathbf{R} = \mathbf{A} \mathbf{A}^H + \sigma_n^2 \mathbf{I}, \quad (8)$$

$$\mathbf{H} = \dot{\mathbf{A}}^H \left[\mathbf{I} - \mathbf{A} (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \right] \dot{\mathbf{A}}. \quad (9)$$

Note that $\mathbf{A}$ is a matrix of all steering vectors, and $\dot{\mathbf{A}}$ is the derivative of $\mathbf{A}$.

• Multiple Signal Classification (MUSIC)

One of the most popular techniques for DoA estimation is the MUSIC algorithm. This method is subspace-based and decomposes the received signal into signal subspace and noise subspace. Therefore, the MUSIC's base is on the covariance matrix of the received signals and their eigenvalue decomposition. Assuming that the covariance matrix $\mathbf{R}_y$ ($N \times N$) has rank N , so $N - K$ eigenvectors correspond to the zero eigenvalue which form the $\mathbf{Q}_n$ matrix. Thus, MUSIC uses a pseudo-spectrum based on eigenvectors. It finds the M largest peaks in the pseudo-spectrum and estimates the direction of signals.

$$P_{MUSIC}(\theta) = \frac{1}{\mathbf{A}(\theta)^H \mathbf{Q}_n \mathbf{Q}_n^H \mathbf{A}(\theta)}, \quad (10)$$

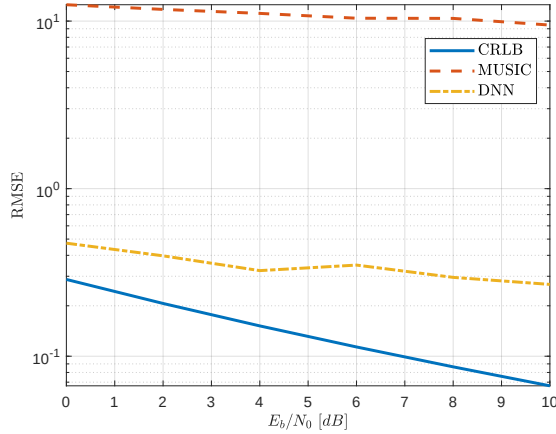


Figure 7: RMSE vs. SNR for 2 target signals and comparison with MUSIC and CRLB in 80% angular range.

where θ is the direction of arrival of the received signal. The eigenvectors matrix $\mathbf{Q}_n$ is orthogonal to the steering vector, $\mathbf{A}(\theta)$, in the signal direction (Gupta, Aditya, and Datta 2016). The denominator would then be zero and, the pseudo-spectrum would be maximized in the desired directions. Noisy conditions with an obvious peak level of the signal are the most conventional usages of the MUSIC method. In these scenarios, the estimation direction has higher accuracy than the similar ones.

Now we want to compare the proposed DNN performance and MUSIC with the CRLB lower bound. In this experiment, we evaluate the proposed DNN's performance in the DoA estimation of two sources in the 80% angular range. At each SNR level, we would calculate RMSE over 10000 Monte Carlo (MC) runs. Additionally, we have also calculated the DoA CRLB of the model. In Fig. 7, we compare the performance of the DOA techniques with each other. In this case, we observe that the DNN-based estimation performs better than the MUSIC and can attain the CRLB. These phenomena can stem from the fact that the MUSIC algorithm's base is on signal subspace decomposition, and it needs a full-rank estimate of the correlation matrix. Therefore, the number of array elements would bound the maximum number of signals it can resolve. Since the DNN-based approach is data-driven, so would not be puzzled by such a problem.

Conclusion

In this paper, we investigate the performance of the DoA estimation using deep neural networks, both in one signal space and two signals space. To do this, we generate a large amount of simulation data. Then, our proposed method utilizes the simulated datasets in training epochs, and the learned multi-layer neural network estimates the direction of arrival signals. The results and evaluation metrics show the performance of our frameworks. Furthermore, we investigate the effect of the antenna array's design parameters on the estimation accuracy of DoA. Results show that an antenna array with more elements has less RMSE. So, it is

more practical to design an antenna array with maximum tolerable antenna elements based on the size and application of the antenna. Further, we investigate the effect of different angular ranges on DoA estimation accuracy and show that the narrower angular range for DoA estimation has much better performance than the full-angular range.

In addition, we compare the performance of our framework and other popular DoA estimation method, MUSIC, and measure the DoA CRLB as the benchmark of our methods. In this comparison, we observe that the proposed DNN-based estimation performs better than other classical methods.

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